Finite Conductor Rings Might not be Quasi-Coherent

Abstract

We resolve an open problem [1] in commutative ring theory and prove that finite conductor rings might not be quasi-coherent. The proof is generated by GPT 5.5 Pro and verified by Claude Opus 4.8 ultracode.

1 Introduction

A ring R is a *finite conductor ring* if the ideals $aR \cap bR$ and $(0 : c)$ are finitely generated for all $a, b, c \in R$, and *quasi-coherent* if $a_1R \cap \cdots \cap a_nR$ and $(0 : c)$ are finitely generated for all $a_1, \ldots, a_n, c \in R$. Taking $n = 2$ shows that every quasi-coherent ring is finite conductor, and it is not immediate whether the converse is true. We show that the converse is not true, i.e. there exists a finite conductor ring that is not quasi-coherent.

2 Main Proof

Theorem 2.1. *There exists a finite-conductor ring that is not quasi-coherent.*

Proof. First we let

$$B = \frac{\mathbb{F}_2[a, b, c, d]}{\big((a, b, c, d)^3,\ ad + bc\big)}.$$

We use the same letters a, b, c, d for their residue classes in B. Let

$$\mathfrak{m} = (a, b, c, d)B,$$

and therefore $\mathfrak{m}^3 = 0$. Define

$$V = \mathbb{F}_2 a + \mathbb{F}_2 b + \mathbb{F}_2 c + \mathbb{F}_2 d$$

and

$$W = \mathfrak{m}^2.$$

We can find an $\mathbb{F}_2$-basis of W as

$$a^2,\ ab,\ ac,\ b^2,\ bc,\ bd,\ c^2,\ cd,\ d^2.$$

Hence, as an $\mathbb{F}_2$-vector space, we have

$$B = \mathbb{F}_2 \cdot 1_B \oplus V \oplus W, \tag{1}$$

which gives a $\mathbb{F}_2$-basis of B:

$$1,\ a,\ b,\ c,\ d,\ a^2,\ ab,\ ac,\ b^2,\ bc,\ bd,\ c^2,\ cd,\ d^2. \tag{2}$$

In particular,

$$\dim_{\mathbb{F}_2} B = 14,$$

and thus B is a finite ring and hence a Noetherian ring. Notice that the multiplication among the three vector-space components satisfies

$$V^2 \subseteq W, \qquad VW = 0, \qquad W^2 = 0. \tag{3}$$

Indeed, all products in VW have degree at least 3, while all products in W^2 have degree at least 4, and all such products vanish in B. The ring B is local with maximal ideal $\mathfrak{m}$. Indeed, an element with nonzero constant term has the form $1+n$, where $n \in \mathfrak{m}$ and $n^3 = 0$. Then

$$(1+n)(1+n+n^2) = 1 + n^3 = 1.$$

Now we let

$$\mathbf{v} = (a, b, c, d) \in B^4$$

and define

$$M = B^4/B\mathbf{v}. \tag{4}$$

Consider the B-linear map

$$\varphi : B \longrightarrow B^4$$

defined by

$$\varphi(r) = r(a, b, c, d).$$

It is not hard to see that its image is $B\mathbf{v}$.

Lemma 2.2. $\ker(\varphi) = W$.

Proof of Lemma 2.2. If $w \in W = \mathfrak{m}^2$, then

$$wa = wb = wc = wd = 0$$

because $\mathfrak{m}^3 = 0$. Therefore $W \subseteq \ker \varphi$. Conversely, suppose that $r \in \ker \varphi$. By the decomposition (1), write

$$r = \lambda + x + w,$$

where

$$\lambda \in \mathbb{F}_2, \qquad x \in V, \qquad w \in W.$$

Since $r(a, b, c, d) = 0$, we have in particular $ra = 0$. Using $Wa = 0$, we obtain

$$0 = ra = \lambda a + xa.$$

The term λa belongs to V, while xa belongs to W because if we write

$$x = \alpha a + \beta b + \gamma c + \delta d,$$

for $\alpha, \beta, \gamma, \delta \in \mathbb{F}_2$, then we have

$$xa = \alpha a^2 + \beta ab + \gamma ac + \delta ad = \alpha a^2 + \beta ab + \gamma ac + \delta bc \in \mathfrak{m}^2.$$

Since the decomposition in (1) is direct, both terms must vanish. Thus $\lambda = 0$ and $xa = 0$. Now the elements a^2, ab, ac, bc are distinct members of the basis (2), and hence are linearly independent over $\mathbb{F}_2$. Therefore

$$\alpha = \beta = \gamma = \delta = 0,$$

which means $x = 0$, so $r = w \in W$. As a result, $\ker(\varphi) = W$. $\square$

Consequently, we have an exact sequence

$$0 \longrightarrow B/W \xrightarrow{\overline{r}\mapsto r(a,b,c,d)} B^4 \longrightarrow M \longrightarrow 0. \tag{5}$$

For an ideal $I \subseteq B$, define

$$(I : \mathfrak{m}) = \{r \in B : r\mathfrak{m} \subseteq I\}.$$

Lemma 2.3. *If $I \subseteq B$ is generated by at most two elements, then*

$$(I : \mathfrak{m}) = I + W. \tag{6}$$

Proof of Lemma 2.3. If $I = B$, the assertion is immediate. We may therefore suppose that I is a proper ideal.

Since B is local, every generator of a proper ideal is a nonunit. We may write

$$I = (x_1, x_2),$$

where a zero second generator is allowed, and we can write

$$x_i = \ell_i + w_i, \qquad \ell_i \in V, \quad w_i \in W.$$

Note that this is possible because I is proper so it cannot contain a unit, which is equivalent to it having a zero constant term. Let

$$L = \mathbb{F}_2\ell_1 + \mathbb{F}_2\ell_2 \subseteq V$$

and put $r = \dim_{\mathbb{F}_2} L \leq 2$. For subspaces $A, D \subseteq V$, write

$$AD = \operatorname{span}_{\mathbb{F}_2}\{ad : a \in A,\ d \in D\} \subseteq W.$$

We first describe $I \cap W$. An arbitrary element of I has the form

$$(\lambda_1 + v_1 + t_1)x_1 + (\lambda_2 + v_2 + t_2)x_2,$$

where $\lambda_i \in \mathbb{F}_2, v_i \in V, t_i \in W$. Using $VW = W^2 = 0$, this element is equal to

$$(\lambda_1\ell_1 + \lambda_2\ell_2) + (\lambda_1 w_1 + \lambda_2 w_2) + (v_1\ell_1 + v_2\ell_2). \tag{7}$$

It belongs to W precisely when its degree-one component vanishes, that is, when

$$\lambda_1\ell_1 + \lambda_2\ell_2 = 0.$$

The vector space of pairs $(\lambda_1, \lambda_2) \in \mathbb{F}_2^2$ satisfying this equation has dimension $2 - r$. It follows that there exists a subspace $E = \{\lambda_1 w_1 + \lambda_2 w_2 : (\lambda_1, \lambda_2) \in K\}$ such that

$$I \cap W = LV + E \tag{8}$$

and

$$\dim_{\mathbb{F}_2} E \leq 2 - r. \tag{9}$$

Now let $z \in (I : \mathfrak{m})$. Write

$$z = \lambda + x + w, \qquad \lambda \in \mathbb{F}_2, \quad x \in V, \quad w \in W.$$

For every $v \in V$, we have $zv = \lambda v + xv$ because $wV = 0$. Since $z\mathfrak{m} \subseteq I$, we have $zv \in I$ for every $v \in V$. The degree-one part of zv is λv, while the degree-one part of I is contained in L.

If $\lambda = 1$, then $V \subseteq L$, which is impossible because

$$\dim_{\mathbb{F}_2} V = 4 \qquad \text{and} \qquad \dim_{\mathbb{F}_2} L \leq 2.$$

Thus we must have $\lambda = 0$, and therefore $zV \subseteq z\mathfrak{m} \subseteq I$. For any $v \in V$,

$$zv = (x + w)v = xv + wv.$$

Now $w \in W = \mathfrak{m}^2$ and $v \in \mathfrak{m}$, so $wv \in \mathfrak{m}^3 = 0$. Hence $zv = xv \in I$. Additionally, $x, v \in V$, so $xv \in V^2 \subseteq W$, which implies

$$xV \subseteq I \cap W = LV + E. \tag{10}$$

Claim 2.4. $x \in L = \mathbb{F}_2\ell_1 + \mathbb{F}_2\ell_2$.

Proof of Claim 2.4. Suppose, toward a contradiction, that $x \notin L$. We let $U = V/L$. The natural multiplication map

$$\operatorname{Sym}^2_{\mathbb{F}_2}(V) \longrightarrow W$$

has kernel generated by the single quadratic relation $ad + bc$. Consequently,

$$W/LV \cong \frac{\operatorname{Sym}^2_{\mathbb{F}_2}(U)}{\mathbb{F}_2\overline{q}}, \tag{11}$$

where $\overline{q}$ is the image of $ad + bc$ in $\operatorname{Sym}^2_{\mathbb{F}_2}(U)$. It is possible that $\overline{q} = 0$.

Let $\overline{x}$ be the nonzero image of x in U. Multiplication by $\overline{x}$ gives an injective linear map

$$U \longrightarrow \operatorname{Sym}^2_{\mathbb{F}_2}(U), \qquad u \longmapsto \overline{x}\,u. \tag{12}$$

Indeed, the symmetric algebra of U is a polynomial ring and hence an integral domain. The image of the map (12) has dimension

$$\dim_{\mathbb{F}_2} U = 4 - r.$$

Passing to the quotient by the at-most-one-dimensional subspace $\mathbb{F}_2\overline{q}$ can reduce the dimension of this image by at most one. Therefore the image of xV in W/LV has dimension at least

$$(4 - r) - 1 = 3 - r. \tag{13}$$

On the other hand, equation (10) says that the image of xV in W/LV is contained in the image of E. By (9), the latter has dimension at most $2 - r$. This is impossible because

$$3 - r > 2 - r.$$

Therefore $x \in L$. $\square$

By Claim 2.4, it follows that
$$z = x + w \in L + W.$$
The image of I in B/W is precisely L, and hence
$$I + W = L + W.$$
Thus
$$(I : \mathfrak{m}) \subseteq I + W.$$
The reverse inclusion is immediate. Indeed, $I\mathfrak{m} \subseteq I$ and $W\mathfrak{m} = 0$. Therefore
$$(I : \mathfrak{m}) = I + W.$$
□

Let $I \subseteq B$ be an ideal generated by at most two elements. Tensoring the exact sequence (5) with B/I gives an exact sequence
$$0 \longrightarrow \operatorname{Tor}_1^B(B/I, M) \longrightarrow B/(I + W) \xrightarrow{\theta_I} (B/I)^4, \tag{14}$$
where
$$\theta_I(\overline{r}) = (\overline{ra}, \overline{rb}, \overline{rc}, \overline{rd}).$$
The kernel of θ_I is
$$\ker \theta_I = \frac{(I : \mathfrak{m})}{I + W}.$$
Indeed, $\theta_I(\overline{r}) = 0$ if and only if
$$ra, rb, rc, rd \in I,$$
which is equivalent to
$$r\mathfrak{m} \subseteq I.$$
By Lemma 2.3, we have $(I : \mathfrak{m}) = I + W$, and therefore
$$\operatorname{Tor}_1^B(B/I, M) = 0. \tag{15}$$
Now tensor the exact sequence
$$0 \longrightarrow I \longrightarrow B \longrightarrow B/I \longrightarrow 0$$
with M. Since $\operatorname{Tor}_1^B(B/I, M) = 0$, the multiplication map
$$I \otimes_B M \longrightarrow M, \qquad r \otimes m \longmapsto rm, \tag{16}$$
is injective. Thus, for every ideal I generated by at most two elements, the canonical multiplication map
$$I \otimes_B M \longrightarrow M$$
is injective.

Lemma 2.5. *For every $x \in B$,*
$$(0 :_M x) = (0 :_B x)M. \tag{17}$$

Proof of Lemma 2.5. Consider the exact sequence

$$0 \longrightarrow (0 :_B x) \longrightarrow B \longrightarrow xB \longrightarrow 0.$$

Tensoring with M gives an exact sequence

$$(0 :_B x) \otimes_B M \longrightarrow M \longrightarrow xB \otimes_B M \longrightarrow 0.$$

The kernel of the map

$$M \longrightarrow xB \otimes_B M, \qquad m \longmapsto x \otimes m,$$

is therefore $(0 :_B x)M$. Since xB is generated by one element, the multiplication map

$$xB \otimes_B M \longrightarrow M$$

is injective by (16). The composite

$$M \longrightarrow xB \otimes_B M \longrightarrow M$$

is multiplication by x. Hence its kernel is $(0 :_M x)$. Therefore

$$(0 :_M x) = (0 :_B x)M.$$

□

Lemma 2.6. *For every $x, y \in B$,*

$$xM \cap yM = (xB \cap yB)M. \tag{18}$$

Proof. The inclusion

$$(xB \cap yB)M \subseteq xM \cap yM$$

is immediate. Conversely, suppose that $z \in xM \cap yM$, then there exist $m, n \in M$ such that

$$z = xm = yn.$$

Let

$$I = xB + yB.$$

The ideal I is generated by at most two elements, so the map

$$I \otimes_B M \longrightarrow M$$

is injective. In $I \otimes_B M$, consider the element

$$x \otimes m + y \otimes n.$$

Its image in M is

$$xm + yn = z + z = 0,$$

because the characteristic is 2. By injectivity,

$$x \otimes m + y \otimes n = 0$$

in $I \otimes_B M$. There is an exact sequence

$$xB \cap yB \longrightarrow xB \oplus yB \longrightarrow xB + yB \longrightarrow 0, \tag{19}$$

where the first map is

$$t \longmapsto (t, t)$$

and the second map is

$$(r, s) \longmapsto r + s.$$

After tensoring with M, exactness shows that

$$(x \otimes m,\ y \otimes n)$$

comes from an element of

$$(xB \cap yB) \otimes_B M.$$

Multiplying its first component in M, we conclude that

$$z = xm \in (xB \cap yB)M.$$

Therefore

$$xM \cap yM = (xB \cap yB)M.$$

□

Define the idealization

$$C = B \ltimes M.$$

Since B and M are finite-dimensional over $\mathbb{F}_2$, the ring C is finite and hence Noetherian. We identify B with the subring

$$B \oplus 0 \subseteq C.$$

For $x \in B$, we have

$$xC = xB \oplus xM. \tag{20}$$

Using (17), we obtain

$$\begin{aligned}(0 :_C x) &= (0 :_B x) \oplus (0 :_M x) \\ &= (0 :_B x) \oplus (0 :_B x)M \\ &= (0 :_B x)C.\end{aligned} \tag{21}$$

Similarly, using (18), we obtain

$$\begin{aligned}xC \cap yC &= (xB \cap yB) \oplus (xM \cap yM) \\ &= (xB \cap yB) \oplus (xB \cap yB)M \\ &= (xB \cap yB)C.\end{aligned} \tag{22}$$

Thus extending from B to C preserves annihilators of elements of B and intersections of two principal ideals generated by elements of B.

Using the basis (2), we have

$$aB = \operatorname{span}_{\mathbb{F}_2}\{a, a^2, ab, ac, bc\}$$

and

$$bB = \mathrm{span}_{\mathbb{F}_2}\{b, ab, b^2, bc, bd\}.$$

Therefore we have

$$aB \cap bB = \mathrm{span}_{\mathbb{F}_2}\{ab, bc\}. \tag{23}$$

The degree-two part of $(a+b)B$ is spanned by

$$a^2 + ab, \qquad ab + b^2, \qquad ac + bc, \qquad bc + bd. \tag{24}$$

Suppose that a linear combination

$$\alpha(a^2 + ab) + \beta(ab + b^2) + \gamma(ac + bc) + \delta(bc + bd)$$

belongs to $\mathrm{span}_{\mathbb{F}_2}\{ab, bc\}$. Comparing the coefficients of a^2, b^2, ac, bd shows that

$$\alpha = \beta = \gamma = \delta = 0.$$

Hence

$$\mathrm{span}_{\mathbb{F}_2}\{ab, bc\} \cap (a+b)B = 0.$$

Therefore

$$aB \cap bB \cap (a+b)B = 0. \tag{25}$$

Now define

$$u = [0,\ ab + b^2,\ 0,\ bc + bd] \in M. \tag{26}$$

We have

$$u = b[0, a+b, 0, c+d], \tag{27}$$

because $b(0, a+b, 0, c+d) = (0, ab + b^2, 0, bc + bd)$. We also have

$$u = (a+b)[0, b, 0, d], \tag{28}$$

because $(a+b)(0, b, 0, d) = (0, ab + b^2, 0, ad + bd)$ and $ad = bc$. Finally,

$$u = a[b, b, d, d]. \tag{29}$$

Indeed,

$$a(b, b, d, d) = (ab, ab, ad, ad) = (ab, ab, bc, bc),$$

and the difference between this vector and the representative in (26) is

$$(ab, b^2, bc, bd) = b(a, b, c, d),$$

which is zero in $M = B^4/B(a, b, c, d)$. Thus

$$u \in aM \cap bM \cap (a+b)M. \tag{30}$$

Claim 2.7. *$u \neq 0$.*

Proof of Claim 2.7. Suppose that $u = 0$. Then there exists $r \in B$ such that

$$(0, ab + b^2, 0, bc + bd) = r(a, b, c, d). \tag{31}$$

Write

$$r = \lambda + \ell + w, \qquad \lambda \in \mathbb{F}_2, \quad \ell \in V, \quad w \in W.$$

The left-hand side of (31) is homogeneous of degree 2. The degree-one part of the right-hand side is $\lambda(a, b, c, d)$, so $\lambda = 0$. Since $W\mathfrak{m} = 0$, equation (31) becomes

$$(0, ab + b^2, 0, bc + bd) = \ell(a, b, c, d).$$

The first coordinate gives

$$\ell a = 0.$$

Write

$$\ell = \alpha a + \beta b + \gamma c + \delta d,$$

then we have

$$\ell a = \alpha a^2 + \beta ab + \gamma ac + \delta bc.$$

The elements a^2, ab, ac, bc are linearly independent. Therefore $\ell = 0$, which contradicts the fact that the second coordinate $ab + b^2$ of the left-hand side of (31) is nonzero. Thus $u \neq 0$. □

By Claim 2.7, we know

$$0 \neq u \in aM \cap bM \cap (a + b)M. \tag{32}$$

Inside $C = B \ltimes M$, the element $(0, u)$ is therefore a nonzero member of

$$aC \cap bC \cap (a + b)C. \tag{33}$$

This contrasts with the equality

$$aB \cap bB \cap (a + b)B = 0.$$

Now we are ready to prove the main theorem. Let

$$C^{(\mathbb{N})} = \left\{ (c_n)_{n \geq 1} \in C^{\mathbb{N}} : c_n = 0 \text{ for all but finitely many } n \right\},$$

and also let

$$\Delta(B) = \{(b, b, b, \ldots) : b \in B\}$$

be the diagonal copy of B in $C^{\mathbb{N}}$. Define

$$R = \Delta(B) + C^{(\mathbb{N})} \subseteq C^{\mathbb{N}}. \tag{34}$$

Thus R consists precisely of those sequences in C that are eventually equal to a fixed element of B. Addition and multiplication in R are coordinatewise. The identity of R is $(1, 1, 1, \ldots)$. For every $n \geq 1$, let $e_n \in R$ be the sequence having 1 in the n-th coordinate and 0 elsewhere.

Lemma 2.8. *R is a finite conductor.*

Proof of Lemma 2.8. Let

$$x = (x_n), \qquad y = (y_n)$$

be elements of R. There exist a finite set $F \subseteq \mathbb{N}$ and elements $x_0, y_0 \in B$ such that

$$x_n = x_0, \qquad y_n = y_0 \qquad \text{for every } n \notin F. \tag{35}$$

Because B is Noetherian, the ideal $x_0B \cap y_0B$ is finitely generated. Choose

$$x_0B \cap y_0B = (d_1, \dots, d_s)_B. \tag{36}$$

By equation (22),

$$x_0C \cap y_0C = (x_0B \cap y_0B)C = (d_1, \dots, d_s)_C. \tag{37}$$

For each i, define $\widetilde{d}_i \in R$ by

$$(\widetilde{d}_i)_n = \begin{cases} 0, & n \in F, \\ d_i, & n \notin F. \end{cases} \tag{38}$$

Each $\widetilde{d}_i$ belongs to $xR \cap yR$. Indeed, choose $p_i, q_i \in B$ such that

$$d_i = x_0p_i = y_0q_i.$$

Define $\widetilde{p}_i, \widetilde{q}_i \in R$ to be zero on F and equal to p_i, q_i, respectively, outside F. Then

$$\widetilde{d}_i = x\widetilde{p}_i = y\widetilde{q}_i.$$

For every $n \in F$, the ideal $x_nC \cap y_nC$ is finitely generated because C is Noetherian. Choose generators

$$g_{n1}, \dots, g_{nt_n}.$$

Regard e_ng_{nj} as the element of R supported only at coordinate n, where its value is g_{nj}.

Claim 2.9. *The finite set*

$$\{\widetilde{d}_1, \dots, \widetilde{d}_s\} \cup \{e_ng_{nj} : n \in F,\ 1 \le j \le t_n\} \tag{39}$$

generates $xR \cap yR$.

Proof of Claim 2.9. Let

$$z = (z_n) \in xR \cap yR,$$

and let $z_0 \in B$ be its eventual value. Then

$$z_0 \in x_0B \cap y_0B.$$

Choose $\beta_1, \dots, \beta_s \in B$ such that

$$z_0 = \sum_{i=1}^{s} d_i\beta_i. \tag{40}$$

For every $n \notin F$, we have

$$z_n \in x_0C \cap y_0C = (d_1, \dots, d_s)C.$$

Therefore there exist $\gamma_{in} \in C$ such that

$$z_n = \sum_{i=1}^{s} d_i \gamma_{in}.$$

Since $z_n = z_0$ for all sufficiently large n, we may choose $\gamma_{in} = \beta_i$ for all sufficiently large n. Consequently, the coefficient sequences

$$r_i = (\gamma_{in})_{n \geq 1},$$

with arbitrary values assigned at the finitely many coordinates in F, belong to R. The element

$$z - \sum_{i=1}^{s} \widetilde{d}_i r_i$$

is supported inside F. At each $n \in F$, its n-th coordinate belongs to $x_n C \cap y_n C$. It is therefore generated by the coordinate-supported elements $e_n g_{nj}$. Thus $xR \cap yR$ is finitely generated. $\square$

We now prove the corresponding statement for annihilators. Let $x = (x_n) \in R$, and suppose that its eventual value is $x_0 \in B$. Choose generators

$$(0 :_B x_0) = (h_1, \dots, h_t)_B.$$

By equation (21),

$$(0 :_C x_0) = (h_1, \dots, h_t)_C.$$

Modified diagonal copies of the h_i, together with generators of the finitely many exceptional annihilators

$$(0 :_C x_n),$$

generate $(0 :_R x)$. Therefore, for every $x, y \in R$, both $(0 :_R x)$ and $xR \cap yR$ are finitely generated. Hence R is a finite-conductor ring. $\square$

Finally, we prove that R is not quasi-coherent, thus completing the proof. Regard $a, b, a+b$ as constant diagonal elements of R, and put

$$J = aR \cap bR \cap (a+b)R.$$

Every element of J has finite support. Indeed, let $z \in J$, and let $z_0 \in B$ be its eventual value. Then

$$z_0 \in aB \cap bB \cap (a+b)B.$$

By (25), we have

$$aB \cap bB \cap (a+b)B = 0.$$

Therefore $z_0 = 0$, so z is eventually zero. For every $n \geq 1$, define

$$u_n = e_n(0, u) \in R.$$

By (32), we get

$$(0, u) \in aC \cap bC \cap (a+b)C.$$

Therefore

$$0 \neq u_n \in J \qquad \text{for every } n \geq 1. \tag{41}$$

Suppose, toward a contradiction, that J is finitely generated. Write

$$J = (z^{(1)}, \ldots, z^{(m)})_R.$$

Each $z^{(i)}$ has finite support. Hence

$$F = \bigcup_{i=1}^{m} \operatorname{supp}(z^{(i)})$$

is finite. Since multiplication in $R \subseteq C^{\mathbb{N}}$ is coordinatewise, multiplication cannot create a nonzero coordinate outside the support of a generator. Therefore every element of

$$(z^{(1)}, \ldots, z^{(m)})R$$

is supported inside F. Choose $n \notin F$. Then $u_n \in J$, but u_n is not supported inside F. This is a contradiction. Therefore

$$aR \cap bR \cap (a+b)R$$

is not finitely generated. Hence R is not quasi-coherent.

$\square$

References

[1] Sarah Glaz. Open problems in commutative ring theory. In Marco Fontana, Sophie Frisch, and Sarah Glaz, editors, *Commutative Algebra: Recent Advances in Commutative Rings, Integer-Valued Polynomials, and Polynomial Functions*, pages 325–340. Springer, New York, 2014. Problem 20.